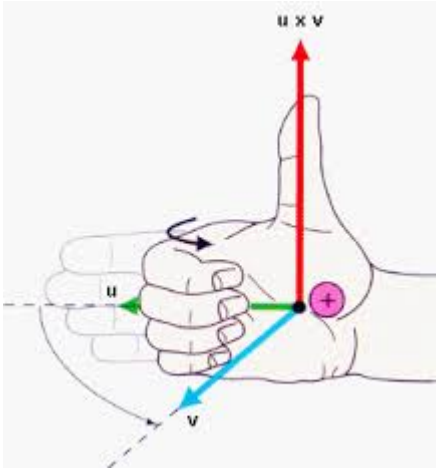
11.4 The Cross Product

- The *cross product* of two vectors is used to describe how a plane is tilting in space.
- If **u** and **v** are non-parallel vectors in space, then they determine a plane. So we can define **u**×**v** to be the vector perpendicular to this plane. (See the definition later.)
- We will use the *right-hand rule*.
- Also called the *vector product*.



Definition **Cross Product**

$$\mathbf{u}\times\mathbf{v}=(|\mathbf{u}||\mathbf{v}|\sin\theta)\mathbf{n}$$

θ =the angle between **u** and **v**

n = the unit vector perpendicular to the plane determined by **u** and **v** according to the right-hand rule

If $\mathbf{u}=\mathbf{0}$ or $\mathbf{v}=\mathbf{0}$, then $\mathbf{u}\times\mathbf{v}=\mathbf{0}$.

If **u** is parallel to **v**, we also have $\mathbf{u}\times\mathbf{v}=\mathbf{0}$.

$\underline{u} \times \underline{v} = \underline{0}$ if & only if $\underline{u} // \underline{v}$ OR $\underline{u} = \underline{0}$ OR $\underline{v} = \underline{0}$

Remark: If $\underline{u} \neq \underline{0}$ & $\underline{v} \neq 0$, then

$\underline{u} \times \underline{v} = \underline{0} \iff \underline{u} // \underline{v}$

$\underline{u} \cdot \underline{v} = 0 \iff \underline{u} \perp \underline{v}$

Properties of the Cross Product

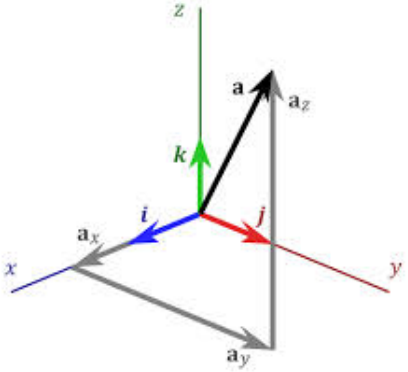
Let $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ be vectors and let r, s be scalars. Then,

1. $(r\mathbf{u})\times(s\mathbf{v}) = (rs)(\mathbf{u}\times\mathbf{v})$
2. $\mathbf{u}\times(\mathbf{v}+\mathbf{w}) = \mathbf{u}\times\mathbf{v} + \mathbf{u}\times\mathbf{w}$
3. $(\mathbf{u}+\mathbf{v})\times\mathbf{w} = \mathbf{u}\times\mathbf{w} + \mathbf{v}\times\mathbf{w}$
4. $\mathbf{v}\times\mathbf{u} = -(\mathbf{u}\times\mathbf{v})$
5. $\mathbf{0}\times\mathbf{u} = \mathbf{0}$
6. $\mathbf{u}\times(\mathbf{v}\times\mathbf{w}) = (\mathbf{u}\cdot\mathbf{w})\mathbf{v} - (\mathbf{u}\cdot\mathbf{v})\mathbf{w}$

Later, as an exercise, to be verified by comparing components.

Remark:

1. In general, $(\mathbf{u}\times\mathbf{v})\times\mathbf{w} \neq \mathbf{u}\times(\mathbf{v}\times\mathbf{w})$. Cross products are not associative.
2. Writing $\mathbf{u}\times\mathbf{v}\times\mathbf{w}$ is meaningless, a pair of parentheses should be included.



$|\underline{i} \times \underline{j}| = |\underline{i}||\underline{j}| \sin\frac{\pi}{2}$

$= 1$

$\underline{i} \times \underline{j}$ is parallel to $\underline{k}$ (by the RHR)

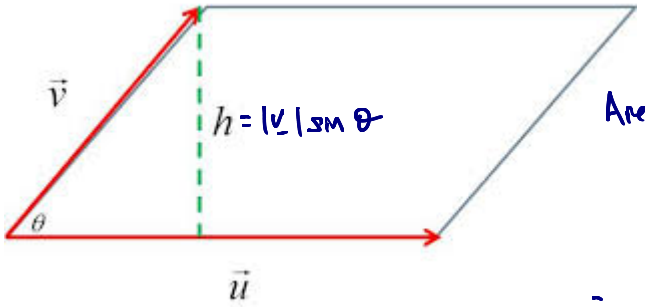
$\Rightarrow \quad \underline{i} \times \underline{j} = \underline{k}, \quad \underline{j} \times \underline{i} = -\underline{k}$

$\underline{j} \times \underline{k} = \underline{i}, \quad \underline{k} \times \underline{j} = -\underline{i}$

$\underline{k} \times \underline{i} = \underline{j} \qquad \underline{i} \times \underline{k} = -\underline{j}$

$\underline{i} \times \underline{i} = \underline{0} \qquad \underline{j} \times \underline{j} = \underline{0} \qquad \underline{k} \times \underline{k} = \underline{0}$

|u×v| is the Area of a Parallelogram



Area of parallelogram =

height × base

$= h \times |\underline{u}|$

$= |\underline{v}| \sin\theta \, |\underline{u}|$

$= |\underline{u}| \, |\underline{v}| \sin\theta$

$= |\underline{u} \times \underline{v}|$

Determinant Formula of u×v

$$\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k} \quad \langle u_1, u_2, u_3 \rangle$$

$$\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k} \quad \langle v_1, v_2, v_3 \rangle$$

$$\mathbf{u}\times\mathbf{v} = (u_1\underline{i} + u_2\underline{j} + u_3\underline{k}) \times (v_1\underline{i} + v_2\underline{j} + v_3\underline{k})$$

$$= u_1v_2(\underline{i}\times\underline{j}) + u_1v_3(\underline{i}\times\underline{k}) + u_2v_1(\underline{j}\times\underline{i}) + u_2v_3(\underline{j}\times\underline{k}) +$$

$$u_3v_1(\underline{k}\times\underline{i}) + u_3v_2(\underline{k}\times\underline{j})$$

($\underline{i}\times\underline{j} = \underline{k}$, $\underline{i}\times\underline{k} = -\underline{j}$, $\underline{j}\times\underline{i} = -\underline{k}$, $\underline{j}\times\underline{k} = \underline{i}$, $\underline{k}\times\underline{i} = \underline{j}$, $\underline{k}\times\underline{j} = -\underline{i}$)

$$= (u_2v_3 - u_3v_2)\underline{i} - (u_1v_3 - u_3v_1)\underline{j} + (u_1v_2 - u_2v_1)\underline{k}$$

$$= \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix}\underline{i} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix}\underline{j} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}\underline{k}$$

$$= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$

determinants

Example: Find the cross products $\mathbf{u}\times\mathbf{v}$ and $\mathbf{v}\times\mathbf{u}$ for

$$\mathbf{u} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$$
$$\mathbf{v} = -4\mathbf{i} + 3\mathbf{j} + \mathbf{k}$$

$$\underline{u} \times \underline{v} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 2 & 1 & 1 \\ -4 & 3 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix} \underline{i} - \begin{vmatrix} 2 & 1 \\ -4 & 1 \end{vmatrix} \underline{j} + \begin{vmatrix} 2 & 1 \\ -4 & 3 \end{vmatrix} \underline{k}$$

$$= -2\underline{i} - 6\underline{j} + 10\underline{k}$$

Example: Find a unit vector perpendicular to the plane containing $P(1,-1,0)$, $Q(2,1,-1)$, $R(-1,1,2)$. Also, find the area of the triangle ΔPQR.

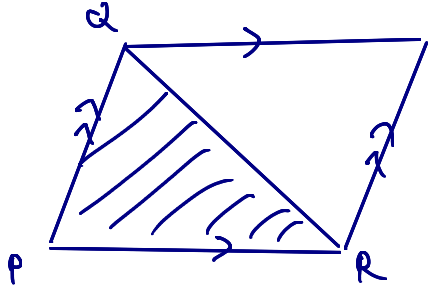
A vector $\perp$ to this plane can be $\vec{PQ} \times \vec{PR}$

$\vec{PQ} = \langle 1, 2, -1 \rangle$ $\quad$ $\vec{PR} = \langle -2, 2, 2 \rangle$

$$\vec{PQ} \times \vec{PR} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & 2 & -1 \\ -2 & 2 & 2 \end{vmatrix} = 6\underline{i} + 6\underline{k}$$

So $6\underline{i} + 6\underline{k}$ is a vector $\perp$ to this plane.

$\therefore$ A unit vector $\perp$ to this plane is $\dfrac{6\underline{i}+6\underline{k}}{|6\underline{i}+6\underline{k}|} = \dfrac{1}{\sqrt{2}}\underline{i} + \dfrac{1}{\sqrt{2}}\underline{k}$



Area of ΔPQR =

$\frac{1}{2}$ (Area of the parallelogram determined by $\vec{PQ}$ & $\vec{PR}$)

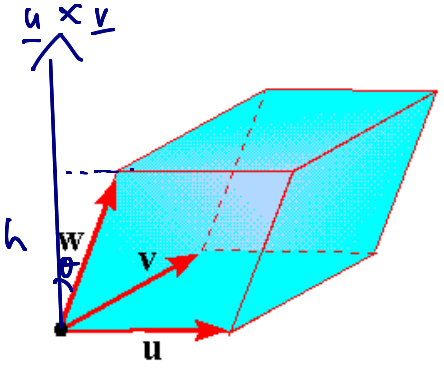
$= \frac{1}{2} |\vec{PQ} \times \vec{PR}| = \frac{1}{2} |6\underline{i} + 6\underline{k}|$

$= \underline{3\sqrt{2}}$

Triple Scalar or Box Product

Recall that $\underline{a} \cdot \underline{b} = |\underline{a}||\underline{b}| \cos\theta$

The product $(\mathbf{u}\times\mathbf{v})\cdot\mathbf{w}$ is called the triple scalar product of **u**, **v**, **w**.



Volume of parallelpiped

= height × base area.

$= h \times |\underline{u} \times \underline{v}|$

$= |\underline{w}| \cos\theta \, |\underline{u} \times \underline{v}|$

$= |\underline{w} \cdot (\underline{u} \times \underline{v})|$

$= |(\underline{u} \times \underline{v}) \cdot \underline{w}|$

$h = |w| \cos\theta$

$= \dfrac{\underline{w} \cdot (\underline{u} \times \underline{v})}{|\underline{u} \times \underline{v}|}$.

Triple Scalar Product as a Determinant

$u = \langle u_1, u_2, u_3 \rangle \quad w = \langle w_1, w_2, w_3 \rangle$

$v = \langle v_1, v_2, v_3 \rangle$

$$(\mathbf{u}\times\mathbf{v}) \cdot \mathbf{w} = \left(\begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \underline{i} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \underline{j} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \underline{k} \right) \cdot (w_1 \underline{i} + w_2 \underline{j} + w_3 \underline{k})$$

$$= \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} w_1 - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} w_2 + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} w_3$$

$$= \begin{vmatrix} w_1 & w_2 & w_3 \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$$

Example: Calculate the volume of the box (parallelepiped) determined by vectors

$\mathbf{u} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$

$\mathbf{v} = -2\mathbf{i} + 3\mathbf{k}$

$\mathbf{w} = 7\mathbf{j} - 4\mathbf{k}$

$$\text{Volume} = |(\underline{u} \times \underline{v}) \cdot \underline{w}|$$

$$= \begin{vmatrix} 1 & 2 & -1 \\ -2 & 0 & 3 \\ 0 & 7 & -4 \end{vmatrix}$$

$$= |1(-21) - 2(8) - 1(-14)|$$

$$= \underline{\underline{23}} \text{ unit}^3$$

Remark:

- **u**, **v**, **w** are coplanar $\Leftrightarrow (\mathbf{u}\times\mathbf{v})\bullet\mathbf{w} = 0$.

  ⇓

  they lie on the same plane

- If any of the two vectors **u**, **v**, **w** are the same, then the triple scalar product is zero. That is,

  $(\mathbf{u}\times\mathbf{u})\bullet\mathbf{v} = 0$, → since $\underline{u}\times\underline{u} = \underline{0}$

  $(\mathbf{u}\times\mathbf{v})\bullet\mathbf{u} = 0$,

  $(\mathbf{u}\times\mathbf{v})\bullet\mathbf{v} = 0$.

  $\underline{u}\times\underline{v}$ ⊥ $\underline{u}$ (& $\underline{v}$)

  and so their dot product is zero.

Example:

Find the volume of the parallelepiped having

$\mathbf{u}=\mathbf{i}+2\mathbf{j}-\mathbf{k}$, $\mathbf{v}=-2\mathbf{j}+3\mathbf{k}$ and $\mathbf{w}=\mathbf{i}+7\mathbf{j}-4\mathbf{k}$ as its sides.

$$\text{Volume} = |(\underline{u}\times\underline{v})\cdot\underline{w}| = \begin{vmatrix} 1 & 2 & -1 \\ 0 & -2 & 3 \\ 1 & 7 & -4 \end{vmatrix}$$

$$= |-13 + 6 - 2| = 9 \text{ units}$$

Example:

Show that the points $P(0, 0, 3)$, $Q(2, -1, 2)$, $R(3, 2, 1)$ and $S(1, 3, 2)$ are coplanar.

We can do this by showing that

$$(\vec{PQ}\times\vec{PR})\cdot\vec{PS} = 0$$

$$\vec{PQ} = \langle 2, -1, -1\rangle \quad \vec{PR} = \langle 3, 2, -2\rangle \quad \vec{PS} = \langle 1, 3, -1\rangle$$

$$\Rightarrow (\vec{PQ}\times\vec{PR})\cdot\vec{PS} = \begin{vmatrix} 2 & -1 & -1 \\ 3 & 2 & -2 \\ 1 & 3 & -1 \end{vmatrix} = 8 - 1 - 7 = 0$$

$\vec{PQ}, \vec{PR}, \vec{PS}$ are coplanar & so P, Q, R, S are coplanar.

Ans: So S: (0, 3, 2) OR (1, 3, 3) OR (1, 3, 1) OR (2, 3, 2)

Example:

Let $P(1, 2, 3)$, $Q(2, 4, 2)$, $R(1, 3, 2)$ and $S(a, 3, b)$ be four points in space and let $\vec{RS}$ be a unit vector. Find all possible coordinates of S if the volume of the parallelepiped determined by the vectors $\vec{PQ}$, $\vec{PR}$ and $\vec{PS}$ is 1 unit.

$\vec{PQ} = \langle 1, 2, -1 \rangle$, $\vec{PR} = \langle 0, 1, -1 \rangle$, $\vec{PS} = \langle a-1, 1, b-3 \rangle$

$\vec{RS} = \langle a-1, 0, b-2 \rangle$

① $\vec{RS}$ is a unit vector $\Rightarrow (a-1)^2 + (b-2)^2 = 1$

② $|(\vec{PQ} \times \vec{PR}) \cdot \vec{PS}| = \begin{vmatrix} 1 & 2 & -1 \\ 0 & 1 & -1 \\ a-1 & 1 & b-3 \end{vmatrix} = |(b-2) - 2(a-1) + (a-1)| = |b-a-1|$

Volume $= 1 \Rightarrow |b-a-1| = 1$

$\Rightarrow b-a-1 = 1$ OR $b-a-1 = -1$

$b = a+2$ OR $b = a$

Use ①: $(a-1)^2 + a^2 = 1$; $(a-1)^2 + (a-2)^2 = 1$

$\Rightarrow 2a^2 - 2a = 0$; $2a^2 - 6a + 5 = 1$

$\Rightarrow a = 0$ OR $a = 1$; $2a^2 - 6a + 4 = 0$

$\Rightarrow b = 2$ OR $b = 3$; $a^2 - 3a + 2 = 0$

$\Rightarrow a = 2$ OR $a = 1$

$b = 2$ OR $b = 1$

Example:

Show that for any vectors **u**, **v** and **w**,

$\mathbf{u}\times(\mathbf{v}\times\mathbf{w})+\mathbf{v}\times(\mathbf{w}\times\mathbf{u})+\mathbf{w}\times(\mathbf{u}\times\mathbf{v})=\mathbf{0}$. — (+)

HINT: The first thing to do is to verify that

(*) $\underline{u}\times(\underline{v}\times\underline{w}) = (\underline{u}\cdot\underline{w})\underline{v} - (\underline{u}\cdot\underline{v})\underline{w}$

by checking that the components are consistent on both sides.

Then apply this (*) to each of the three terms in (+)